

$$1) Z = \frac{V_{ef}}{I_{ef}} = \frac{100V}{0,2A} = \boxed{500 \Omega} \quad (1)$$

$$b) \cos \phi = \frac{R}{Z} \rightarrow \cos \frac{\pi}{3} = \frac{R}{500 \Omega} = \frac{1}{2} \rightarrow \boxed{R = 250 \Omega}$$

$$\frac{X_C}{R} = \tan \phi = \tan \left(-\frac{\pi}{3}\right) \rightarrow X_C = \sqrt{3} \times 250 \Omega = 433,01 \Omega$$

$$\rightarrow 433,01 \Omega = \frac{1}{\omega C} = \frac{1}{2\pi f C} = \frac{1}{100\pi \frac{1}{5} C} \rightarrow \boxed{C = 7,35 \mu F}$$

$$c) V_{R,ef} = R I_{ef} = 250 \Omega \times 0,2A \rightarrow \boxed{V_{R,ef} = 50V}$$

$$d) V_{C,ef} = X_C I_{ef} = 433,01 \Omega \times 0,2A \rightarrow \boxed{V_{C,ef} = 86,6V}$$

$$e) X_L = X_C \rightarrow \omega L = \frac{1}{\omega C} \rightarrow L = \frac{1}{\omega^2 C} = \frac{1}{(2\pi f)^2 C}$$

$$\rightarrow L = \frac{1}{4\pi^2 \times \left(100\frac{1}{5}\right)^2 \times 7,35 \times 10^{-6} F} \rightarrow \boxed{L = 344,6 mH}$$

$$f) P = R I_{ef}^2 = 250 \Omega \times (0,2A)^2 \rightarrow \boxed{P = 10W}$$

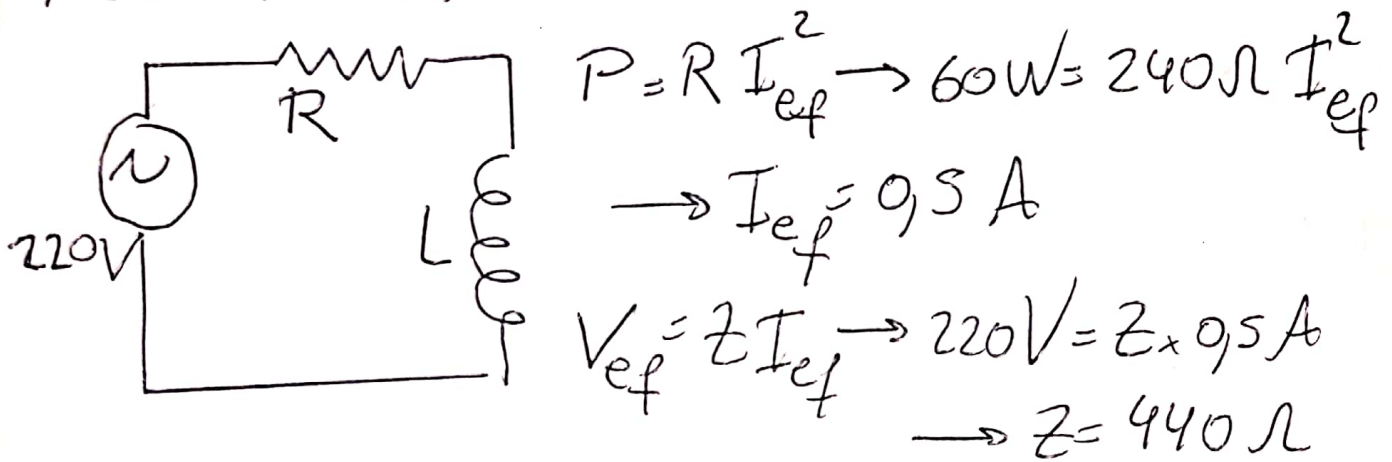
g) EN PARALELO CADA UNO A 100V EFICACES

h) RC LA CORRIENTE ADELANTA $\frac{\pi}{3}$ A LA TENSION

RLC, HAY RESONANCIA Y LA CORRIENTE Y LA TENSION ESTÁN EN FASE.

$$3) \begin{cases} 120V = RI \\ 60W = RI^2 \end{cases} \rightarrow 60W = 120V \times I \rightarrow I = 0,5A \quad (2) \\ \rightarrow \underline{R = 240\Omega}$$

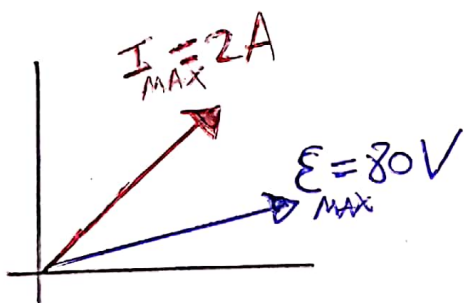
RED DOMICILIARIA:



$$Z^2 = R^2 + X_L^2 \rightarrow (440\Omega)^2 = (240\Omega)^2 + X_L^2 \rightarrow X_L = 368,8\Omega$$

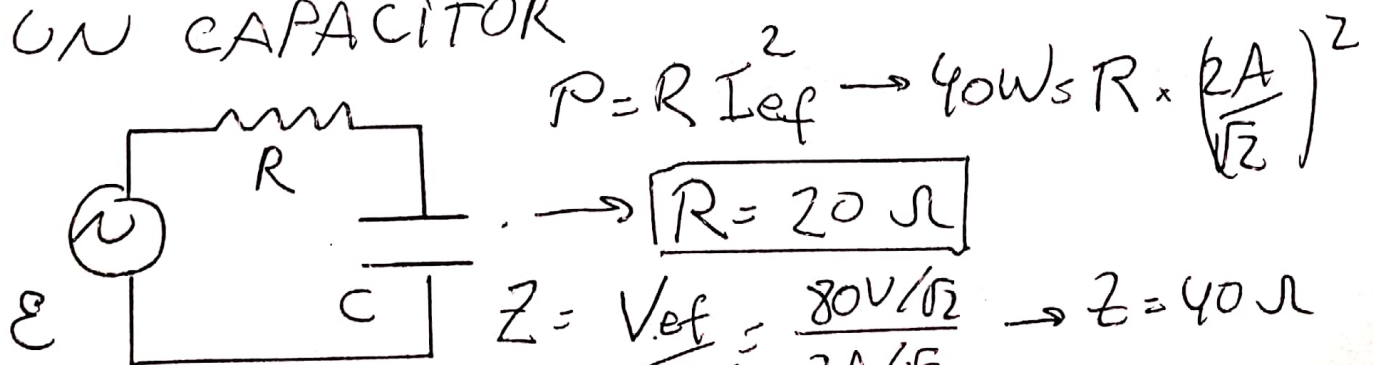
$$X_L = \omega L = 2\pi f L \rightarrow 368,78\Omega = 100\pi \frac{1}{s} L \rightarrow \underline{L = 1,17H}$$

4)



COMO HAY DISIPACIÓN DE POTENCIA HAY UNA RESISTENCIA. Y COMO LA

CORRIENTE ADELANTA A LA TENSION HAY UN CAPACITOR



$$Z^2 = R^2 + X_C^2 \rightarrow (40\Omega)^2 = (20\Omega)^2 + X_C^2 \rightarrow X_C = 34,64\Omega$$

$$X_C = \frac{1}{\omega C} \rightarrow C = \frac{1}{2\pi f X_C} = \frac{1}{100\pi \frac{1}{s} \times 34,64\Omega} \rightarrow \underline{C = 91,9\mu F}$$

5) LA CORRIENTE ATRASA $\rightarrow$ INDUCTIVO. (3)

b) $X_L > X_C$, PARA RESONANCIA DEBO AUMENTAR X_C , PERO $X_C = \frac{1}{\omega C} \rightarrow$ DEBO REDUCIR C

$$a) P = R I_{ef}^2 \rightarrow 80W = R \left(\frac{2A}{\sqrt{2}} \right)^2 \rightarrow R = 40 \Omega$$

$$Z = \frac{E_0}{I_0} = 50 \Omega \quad Z^2 = R^2 + (X_L - X_C)^2 \rightarrow$$
$$\rightarrow (50\Omega)^2 = (40\Omega)^2 + (X_L - X_C)^2$$
$$\rightarrow \boxed{X_L - X_C = 30 \Omega}$$

$$6) a) I_{ef} = \frac{0,4A}{\sqrt{2}} = 0,283 A \quad \underline{X_L > X_C}$$

$$S = V_{ef} I_{ef} \rightarrow 60VA = V_{ef} \times 0,283A \rightarrow V_{ef} = 212,13V$$

$$Z = \frac{V_{ef}}{I_{ef}} = \frac{212,13V}{0,283A} \rightarrow Z = 749,58 \Omega \quad \phi = 37^\circ$$

$$\cos \phi = \frac{R}{Z} \rightarrow R = 749,58 \Omega \times \cos 37^\circ \rightarrow \boxed{R = 598,6 \Omega}$$

$$b) \tan \phi = \frac{X_L - X_C}{R} \rightarrow X_L - X_C = 598,6 \Omega \tan 37^\circ$$

$$\rightarrow \omega L - \frac{1}{\omega C} = 451,08 \Omega$$

(4)

$$\rightarrow 2\pi f L - \frac{1}{2\pi f C} = 451,08 \Omega \rightarrow$$

$$\rightarrow 100\pi \frac{1}{5} L - \frac{1}{100\pi \frac{1}{5} C} = 451,08 \Omega \quad (1)$$

ADÉMÁS RESUENA A 40 Hz!

$$LC = \frac{1}{\omega_r^2} = \frac{1}{(2\pi \times 40 \frac{1}{5})^2} \rightarrow \text{~~XXXXXXXXXX~~}$$

$$\rightarrow LC = 1,583 \times 10^{-5} \text{ s}^2 \quad (2)$$

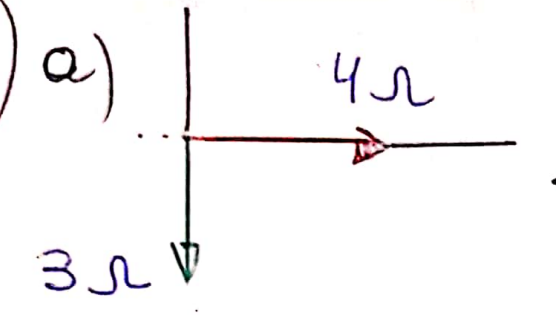
REEMPLAZANDO (2) EN (1):

$$100\pi \frac{1}{5} L - \frac{L}{100\pi \frac{1}{5} \times 1,583 \times 10^{-5} \text{ s}^2} = 451,08 \Omega$$

$$100\pi \frac{1}{5} L - 64\pi \frac{1}{5} L = 451,08 \Omega$$

$$36\pi \frac{1}{5} L = 451,08 \Omega$$

$L = 4 \text{ H}$
$C = 4 \mu\text{F}$

8) a)  $E(t) = 10V \sin(500 \frac{1}{5} t)$ (5)
 $\rightarrow E_{MAX} = 10V \quad R = 4\Omega$
 $\omega = 500 \frac{1}{5} \quad C = 200 \mu F$

$$Z = \sqrt{(4\Omega)^2 + (3\Omega)^2} \rightarrow Z = 5\Omega$$

$$I_{ef} = \frac{E_{ef}}{Z} = \frac{10V/\sqrt{2}}{5\Omega} \rightarrow I_{ef} = \sqrt{2} A$$

$$S = V_{ef} I_{ef} = \frac{10V}{\sqrt{2}} \times \sqrt{2} A \rightarrow \boxed{S = 10 VA}$$

$$\cos \phi = \frac{R}{Z} = \frac{4}{5} \quad P = S \cos \phi = 10 \times \frac{4}{5} W \rightarrow \boxed{P = 8 W}$$

b) $\cos \phi = \frac{4}{5} \rightarrow \phi = -36,87^\circ$ CAPACITIVO
 $X_C > X_L$

$$X_C = \frac{1}{\omega C} = \frac{1}{500 \frac{1}{5} \times 200 \times 10^{-6} F} \rightarrow \text{scribbled out}$$

$$\rightarrow X_C = 10 \Omega$$

$$X_C > X_L, \quad X_C - X_L = 3\Omega \rightarrow X_L = 7\Omega$$

$$V_{R, MAX} = R I_{MAX} = 4\Omega \times 2A = 8V$$

$$V_{L, MAX} = X_L I_{MAX} = 7\Omega \times 2A = 14V$$

$$V_{C, MAX} = X_C I_{MAX} = 10\Omega \times 2A = 20V$$

LA TENSION ES $\varepsilon(t) = 10V \sin(500 \frac{1}{s} t)$, y

LA CORRIENTE ADELANTA $36,87^\circ$. LA TENSION EN LA RESISTENCIA VA EN FASE CON LA CORRIENTE, LV 660:

$$V_R(t) = V_{R,MAX} \sin(500 \frac{1}{s} t + 36,87^\circ) \rightarrow$$

$$\rightarrow V_R(t) = 8V \sin(500 \frac{1}{s} t + 36,87^\circ)$$

LA TENSION EN EL CAPACITOR VA 90° ATRAS DE LA TENSION EN LA RESISTENCIA:

$$V_C(t) = V_{C,MAX} \sin(500 \frac{1}{s} t + 36,87^\circ - 90^\circ)$$

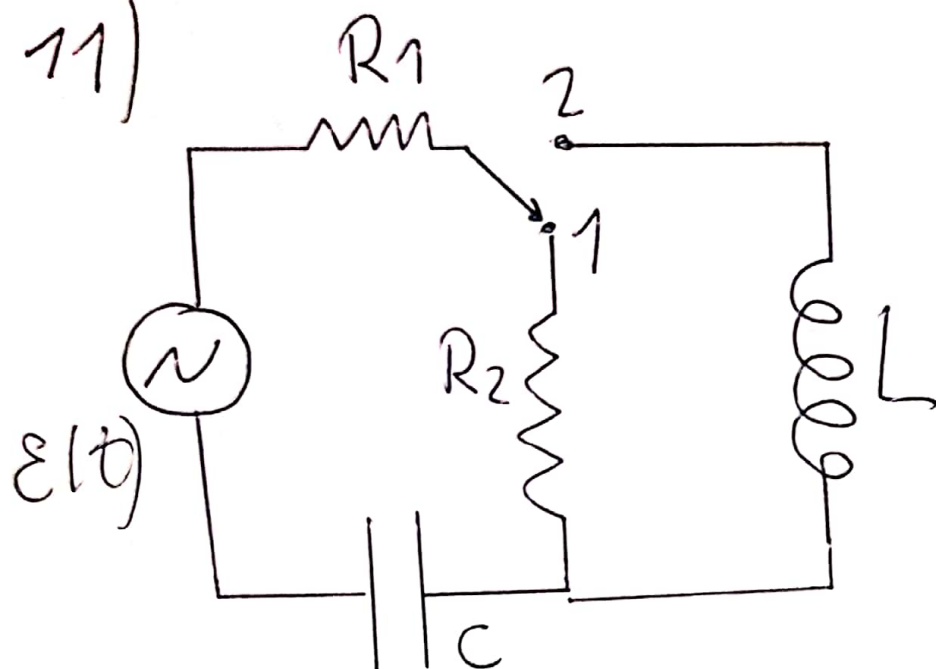
$$\rightarrow V_C(t) = 20V \sin(500 \frac{1}{s} t - 53,13^\circ)$$

LA TENSION EN LA BOBINA ADELANTA 90° A LA TENSION EN LA RESISTENCIA:

$$V_L(t) = 14V \sin(500 \frac{1}{s} t + 126,87^\circ)$$

(7)

11)



a) EN LA POSICIÓN 2 HAY RESONANCIA

$$\rightarrow L = \frac{1}{\omega^2 C} = \frac{1}{\left(400 \frac{1}{s}\right)^2 \times 500 \times 10^{-6} F}$$

$$\boxed{L = 12,5 \text{ mH}}$$

$$b) X_C = \frac{1}{\omega C} = \frac{1}{400 \frac{1}{s} \times 500 \times 10^{-6} F} \rightarrow X_C = 5 \Omega$$

$$Z = \sqrt{(R_1 + R_2)^2 + X_C^2} = \sqrt{(20 \Omega)^2 + (5 \Omega)^2} = 20,62 \Omega$$

$$\cos \phi = \frac{R_1 + R_2}{Z} = \frac{20}{20,62} \rightarrow \boxed{\cos \phi = 0,97}$$